A5 PySpark performance

Lazy evaluation, built-ins over UDFs, joins, partitioning, caching, explain plans.

A5-L1 Lazy evaluation, and why every count() re-runs the plan

Four questions, one scan. The strong version reads the delivery once, however many checks you add.

Transformation or action?	Transformations (filter, select, join, groupBy().agg) build a plan. Actions (count, collect, show, write) run it.
What happens on the second count() of an uncached DataFrame?	The whole plan runs again from the source.
Many checks, one scan	One df.agg(...) with count, count_if, sum, min, max expressions.
Reusing an expensive result on serverless	Write it to a Delta table. cache() is not supported on serverless compute.

A5-L2 Built-in functions over Python UDFs

The strong version runs in the JVM, survives NULLs and reads as a rule. Functions that return Column expressions give you the reuse that UDFs promised.

Why are Python UDFs slow?	Every value crosses from the JVM to a Python worker and back. The plan shows BatchEvalPython.
Measured cost on 2 million strings	Built-in 0.30 s, pandas UDF 1.16 s, Python UDF 1.82 s (PySpark 4.0.4, local).
What does a Python UDF receive for NULL?	None. s.strip() then raises and fails the job. Built-ins return NULL for NULL.
Reusable logic without a UDF	A Python function that takes and returns Column expressions, built from F.when, F.regexp_replace and friends.

A5-L3 Joins, broadcast, partitions and reading the plan

One broadcast join replaces a collect, a dict and a UDF, keeps the work distributed, and shows up clearly in the plan.

Join strategy for a large table and a small lookup	Broadcast the lookup: F.broadcast(small). The plan shows BroadcastHashJoin.
Default auto-broadcast threshold in Spark 4.0	10 MB (spark.sql.autoBroadcastJoinThreshold). AQE can also switch to broadcast at run time.
repartition versus coalesce	repartition(n) shuffles to n partitions. coalesce(n) merges without a shuffle and only reduces.
How do you confirm a filter reached the file scan?	df.explain(mode="formatted") and look for PushedFilters on the scan.